

REVIEW

Intrarenal Determinants of GFR Decline During Loop Diuretic Therapy

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ABSTRACT: Loop diuretics are first-line therapy for hypervolemia. They block the sodium-potassium-2-chloride cotransporter type 1 on the renal afferent arteriole that mediates vasoconstriction and the sodium-potassium-2-chloride cotransporter type 2 on the macula densa cells that mediates the tubuloglomerular feedback response and raises intrarenal pressure that inhibits the myogenic response. These should reduce afferent arteriolar vasoconstriction and increase the glomerular filtration rate; however, more often, the glomerular filtration rate is reduced. This has been attributed to diuretic-induced extracellular volume depletion that activates a systemic neurohormonal response. Accordingly, a fall in glomerular filtration rate can lead to strategies to restore body fluid volumes while reducing the diuretic dosage. Here, we review the regulation of renal hemodynamics during loop diuretic therapy. We propose that a passive, obstructive component of reduced renal blood flow combines with activation of renal afferent nerves, macula densa–derived renin release with intrarenal Ang II (angiotensin II) formation, and generation of vasoconstrictor prostaglandins and thromboxane to provide intrarenal mechanisms of passive and active increases in renal vascular resistance that together can reduce the glomerular filtration rate. The hypothesis that a reduction in the glomerular filtration rate with loop diuretics can derive from intrarenal mechanisms of vasoconstriction rather than from volume depletion suggests novel approaches to manage diuretic resistance and worsening renal function in heart failure.

Key Words: clinical relevance ■ extracellular fluid ■ furosemide ■ renin ■ torsemide

Loop diuretics are used widely to treat hypervolemia in heart failure (HF), but treatment may be complicated by diuretic resistance and worsening renal function (WRF).^{1,2} This is especially frequent in those receiving high-dose diuretics.³ Loop diuretics inhibit the NKCC1 (sodium-potassium-2-chloride cotransporter type 1) on the renal afferent arteriole and thereby reduce afferent arteriolar constriction.⁴ They block NKCC2 (sodium-potassium-2-chloride cotransporter type 2) in macula densa cells, which suppresses the vasoconstrictive tubuloglomerular feedback (TGF) response and thereby also reduces afferent arteriolar constriction.⁵ Furthermore, they increase intratubular fluid volume and raise intrarenal tissue pressure, which reduces the transmural pressure across renal arterioles and thereby diminishes the stimulus for myogenic vasoconstriction.⁶ All of these should increase renal blood flow (RBF), whereas more often both the RBF and glomerular filtration rate (GFR) are reduced.⁷ This has been attributed to the systemic

effects of diuretic-induced extracellular fluid volume depletion activating the sympathetic nervous system and the renal efferent nerve activity to cause renal vasoconstriction and renin release. The outcome is a systemic activation of neurohumoral mechanisms that increase the renal vascular resistance (RVR) and reduce GFR. If this conventional view that volume depletion drives the fall in GFR is correct, then the logical management strategy would be liberalized fluid and salt intake and reduced diuretic dose. However, not only are these strategies counterproductive to the goal of reducing body fluid volumes, but they may also not be effective because a fall in RBF and GFR in rats given furosemide is not prevented by IV infusion of saline to maintain fluid balance, and neurohormonal activation with furosemide may precede natriuresis and volume loss.^{8–10} Regulation of renal hemodynamics by loop diuretics is complex but of great clinical relevance because a falling GFR may interrupt needed volume-depleting therapy in critically ill patients

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Nonstandard Abbreviations and Acronyms

Ang II	angiotensin II
AT1	angiotensin type 1
GFR	glomerular filtration rate
LpA	glomerular ultrafiltration coefficient
NKCC1	sodium-potassium-2-chloride cotransporter type 1
NKCC2	sodium-potassium-2-chloride cotransporter type 2
RBF	renal blood flow
RVR	renal vascular resistance
TGF	tubuloglomerular feedback
TP-Rs	thromboxane prostanoid receptors
TRPV-1	transient receptor potential vanilloid type 1
TxA₂	thromboxane A2
WRF	worsening renal function

and lead to the discontinuation of guideline-directed medical therapy in patients with chronic HF or chronic kidney disease.⁷

Here, we present a novel hypothesis that a fall in RBF and GFR in patients receiving loop diuretics for fluid overload may not originate primarily from a renal response to extracellular fluid volume depletion but rather from events within the kidney itself that are a direct consequence of the actions of the diuretic on the tubules. Inhibition of tubular fluid reabsorption in the thick ascending limb of the loop of Henle raises the volume of intratubular fluid in the distal nephron up to 10-fold.^{11,12} Because the renal capsule and stroma resist swelling, this rise in volume is transmitted as an increase in pressure throughout the kidneys.^{12–16} This could compress and collapse the intrarenal tubules, blood vessels, and capillaries, leading to diuretic resistance and a passive reduction in the RBF and GFR.^{4,8,17} Raised intrarenal pressure can also activate renal baroreceptor afferent nerves to enhance vasoconstriction by an intrarenal mechanism analogous to an axon reflex.^{8,10,18} Moreover, the intrarenal expression of angiotensinogen is increased in rats with experimental HF.¹⁹ An increase in intrarenal Ang II (angiotensin II) in HF exerts adverse effects on renal hemodynamics and diuretic responsiveness. This is contrary to the concept that systemic neurohumoral activation with loop diuretics drives diuretic resistance and WRF. Moreover, plasma renin activity may not be raised in chronic HF.^{19,20} Renin-angiotensin system inhibitors more often cause WRF that is usually attributed to changes in intrarenal hemodynamics and generally reduce the natriuretic response to a loop diuretic.^{21,22} However, loop diuretics block NKCC2 on macula densa cells to increase intrarenal renin release. They also increase renal generation of vasoconstrictor prostaglandins and thromboxane A2

(TxA₂) that contribute to a rise in RVR.^{23,24} This leads to the hypothesis that loop diuretics activate intrarenal factors that elicit a contribution of passive and active increases in RVR. In this view, the fall in GFR during loop diuretic therapy can be initiated primarily by intrarenal events rather than a systemic response to volume depletion, as illustrated in the Figure.

GFR and Renal Hemodynamics

The GFR in normal human subjects or patients with HF declines by ≈25% over 24 hours after administration of a loop diuretic and typically persists over at least a week of loop diuretic therapy.^{25–27} The responses of patients with volume expansion are similar to those of normal subjects, which argues against volume depletion as the sole driver of GFR decline.²⁸ However, renal clearance studies report that GFR and RBF increase >30% over 1 hour after a loop diuretic but decline rapidly and are reduced below baseline from 2 to 24 hours.^{26,28} On the other hand, RBF measured directly in rats by an electromagnetic flow meter or by Doppler velocimeters shows only a progressive fall after loop diuretics even during the first few minutes of diuretic infusion, and the renal cortical and medullary blood flow of normal subjects measured by magnetic resonance imaging falls within 10 minutes of administration of furosemide.^{5,8,10,29} Although the long-term effects of loop diuretics require further study, a reduction in the dose of furosemide over 3 weeks in patients with stable chronic HF and chronic kidney disease increases the GFR by 20%, suggesting a persistent effect to reduce the GFR.^{30,31}

Thus, the early increases in RBF with loop diuretics that have been reported by clearance methods are not confirmed by direct measurement. Renal clearance measurements assume a steady state between filtration and urinary excretion. When urine flow increases abruptly with loop diuretics, this assumption fails. Before diuresis begins, the filtered marker accumulates and becomes concentrated in the tubular fluid to be washed out by the subsequent surge in urine flow. The urinary excretion temporarily exceeds filtration, and the calculated GFR is therefore increased artifactually. Collectively, loop diuretics generally reduce the true RBF and GFR of rodents and humans.

Determination of Glomerular Filtration

The GFR is determined by the difference between the hydraulic and oncotic pressures across the glomerular basement membrane, the glomerular plasma flow, and the glomerular ultrafiltration coefficient (LpA). The net driving force for ultrafiltration of about 15 mmHg is similar to the 10 to 15 mmHg increase in the hydrostatic pressure in the proximal tubule during furosemide infusion into rats.⁸ This could equal the net glomerular

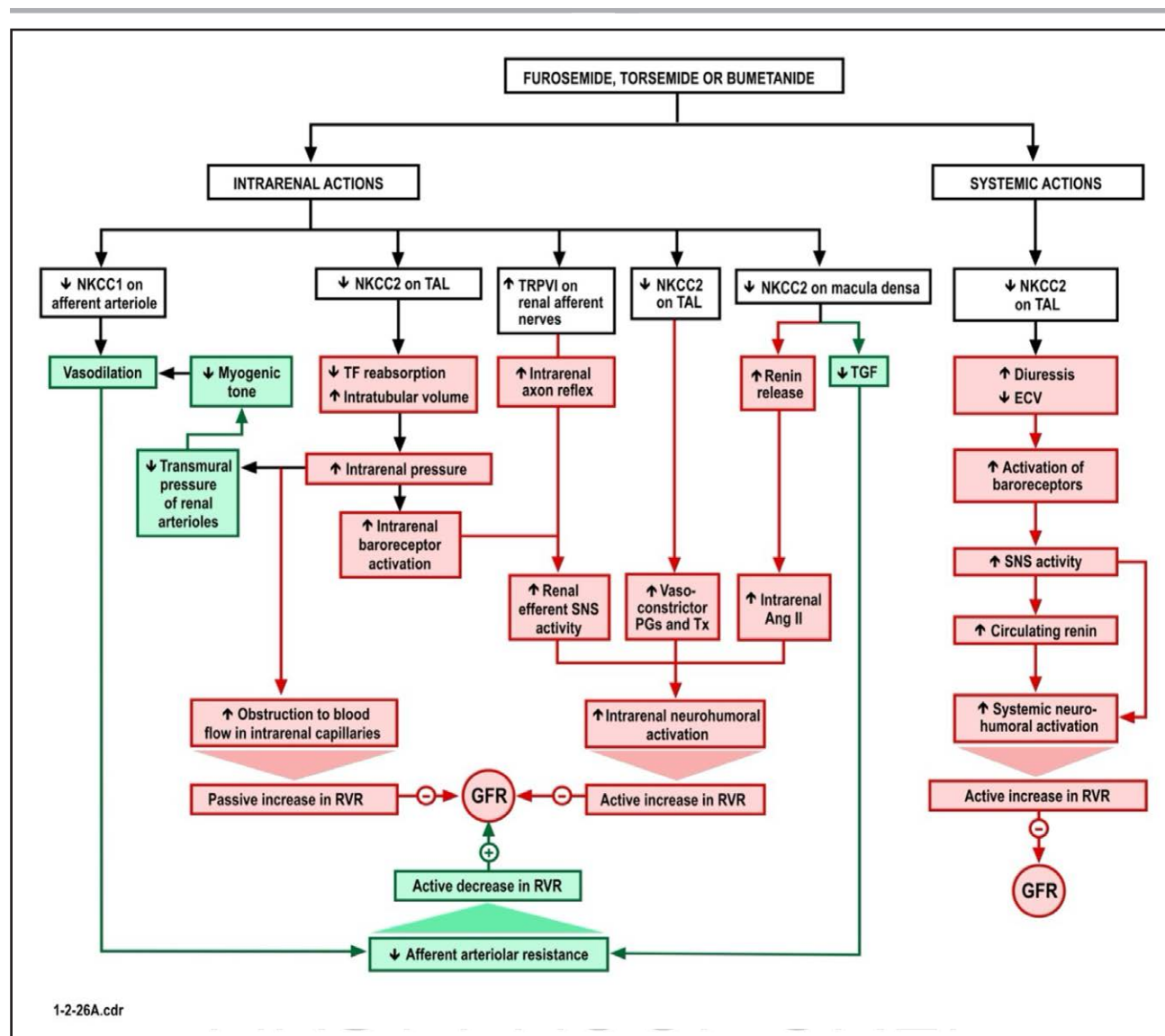


Figure. Diagrammatic representation of intrarenal and systemic mechanisms that can increase or decrease the renal vascular resistance (RVR) and glomerular filtration rate (GFR) during loop diuretic therapy.

Ang II indicates angiotensin II; ECV, extracellular fluid volume; NKCC1, sodium-potassium-2-chloride cotransporter 1; NKCC2, sodium-potassium-2-chloride cotransporter 2; PG, prostaglandin; SNS, sympathetic nervous system; TAL, thick ascending limb of the loop of Henle; TGF, tubuloglomerular feedback; TRPV-1, transient receptor potential vanilloid channel type 1; and Tx, thromboxane.

ultrafiltration pressure at the efferent end of the glomerular capillary, thereby shortening the effective filtration length and reducing filtration substantially. However, the GFR and RBF of rats infused with furosemide fall in parallel, with a marked rise in RVR, indicating that the fall in GFR is due primarily to impaired glomerular plasma flow rather than increased resistance at the filtration barrier.⁸ Micropuncture studies assign the fall in GFR to reductions in the glomerular capillary plasma flow and LpA, similar to that observed with increased intrarenal pressure from venous obstruction.^{15,32} Thus, loop diuretics reduce GFR primarily through mechanisms that increase RVR and reduce renal plasma flow and LpA, reflecting combined vascular and permeability effects rather than simple backpressure on filtration.

Renal Vasodilation With Loop Diuretics

There are 3 mechanisms of renal vasodilation that can be engaged by loop diuretics. First is their inhibition of sodium/chloride reabsorption via NKCC2 at macula densa cells that activates the TGF. This blocks vasoconstriction, predominantly of the afferent arteriole. Second is their inhibition of NKCC1. This blocks direct renal afferent arteriolar vasoconstriction.⁴ Third is their increase in intrarenal pressure, thereby lessening the transmural pressure gradient across intrarenal blood vessels.⁶ This reduces myogenic contraction.⁶

Loop diuretics inhibit TGF even while RBF and GFR decline, implying that concurrent vasoconstrictor influences can override the effects of TGF inhibition

and dominate the net renal hemodynamic response.⁵ A recent study in mice confirms that furosemide suppresses TGF, but this was associated with tubular injury, suggesting that intact TGF is required for tubular integrity, although the clinical relevance for diuretic therapy remains untested.³³

Increase in RVR-Passive Mechanisms

There is an increasing emphasis in HF on the role of increased intrarenal pressure and compression of renal vascular and tubular structures rather than reduced cardiac output as the principal drivers of WRF and adverse outcomes.^{30,34,35} During renal vein constriction in volume-expanded dogs, RBF and GFR are reduced in proportion to the rise in renal interstitial pressure.³⁶

Furosemide administration to rats increases the volume of the distal tubular fluid >10-fold.^{11,12} Because the thin limbs of the loop of Henle and the collecting ducts are the major sites of hydrodynamic resistance to tubular fluid flow, inhibition of fluid resorption between these 2 sites by loop diuretics can account for the increase in intratubular fluid volume in the distal nephron.³⁷ The kidney capsule and stroma are relatively inelastic and thereby limit the swelling of renal tissues during increases in intratubular fluid volume.^{13,14} Therefore, an increase in the fluid volume of the distal nephron with loop diuretics could increase pressures throughout the kidney sufficient to compress and collapse renal tubules, peritubular capillaries, and renal veins, causing a passive increase in RVR, although compression of glomerular capillaries could also account for a decrease in LpA.⁴

Furosemide administration to rats reduces renal cortical and medullary blood flow rapidly. It increases the water content throughout the kidney, affecting both the intracellular and extracellular compartments, reflecting severe renal swelling.³⁸ Studies with renal ultrasound in human subjects reveal that furosemide increases the size of the kidneys acutely by 9% to 23%.³⁹ An increase in the size of the encapsulated kidney should be accompanied by an increase in pressure. Indeed, Doppler studies of intrarenal veins reveal a discontinuous pattern of diastolic flow in patients with severe HF and renal congestion reflecting compression and collapse of small veins. The use of loop diuretics, but not other therapies, is strongly associated with this discontinuous pattern and is normalized by successful decongestion.^{40,41}

Active Renal Vasoconstriction-Intrarenal Renin-Angiotensin System

Inhibition of NKCC2 on macula densa cells by loop diuretics stimulates renin release into the renal interstitium, where it increases intrarenal Ang II, as reflected in

renal lymph.⁴² This causes vasoconstriction of efferent >afferent arterioles that reduces RBF>GFR, although a reduction in LpA contributes further to a fall in GFR.⁴³ Renal Ang II is increased with intrarenal pressure from renal vein constriction⁴⁴ and contributes to the rise in RVR,⁴⁵ suggesting that intrarenal Ang II contributes to increased renal vasoconstriction from renal swelling induced by loop diuretics. Intrarenal Ang II in rat models also generates oxidative stress that activates renal afferent nerves, which in turn enhance efferent sympathetic drive and renal and cardiac damage.^{46,47} Thus, increased Ang II during loop diuretics can derive from several intrarenal mechanisms. It contributes about 40% to the rise in RVR with furosemide in rats and mice, with a smaller effect on GFR.^{4,8} This is consistent with the effect of low-dose infusion of Ang II to reduce RBF>GFR.⁴⁸

Active Renal Vasoconstriction-Renal Nerve Activity

Renal afferent nerve activity from renal baroreceptors is increased by intrarenal or intrapelvic pressure, while that from renal chemoreceptors is increased by the salt concentration or acidity of the pelvic urine or by an increase in renal oxidative stress.⁴⁹ An increase in renal afferent nerve activity generally increases the renal efferent sympathetic nervous system activity and contributes to hypertension, renal vasoconstriction, and renal damage.^{46,47,50–53} Thus, the increases in intrarenal pressure and urinary salt concentration caused by loop diuretics could drive the sharp rise in renal afferent nerve activity that has been recorded and thereby contribute to the rise in efferent nerve activity and RVR.^{54,55}

Renal nerve deafferentation, but not renal nerve denervation, prevents the fall in GFR with furosemide infusion in rats.^{8,10} Thus, furosemide may activate a local intrarenal afferent nerve axon reflex among the renal sympathetic and sensory nerves that are closely related in the pelvic wall.^{10,14,17,50,56} Moreover, many renal afferent nerves express the TRPV-1 (transient receptor potential vanilloid type-1) receptor that can be activated by pressure or chemical stimuli or directly by low concentrations of furosemide. This could contribute to a fall in GFR.^{54,56} Loop diuretics also increase TxA₂ and isoprostanes that activate TP-Rs (thromboxane prostanoic acid receptors) linked to TRPV-1 channels.^{23,54} Thus, loop diuretics could activate TRPV-1 channels to drive renal afferent nerve activity and reflex increases in RVR, but this requires further study.

Intrarenal Prostaglandin Systems

Furosemide increases the excretion of prostaglandins and TxA₂ that contribute to its effects on renin

secretion and renal hemodynamics.^{4,23} Blockade of TxA_2 synthase in patients with HF receiving loop diuretics increases the GFR and RPF and improves diuretic efficacy.⁵⁷

Time-Course of Changes in GFR

Furosemide infusion reduces the GFR and RBF of rats progressively over 1 hour, independent of volume loss. The P_{PT} rises rapidly within the first 20 minutes and declines somewhat thereafter despite maintained diuresis, suggesting that the compression of the kidneys by the renal parenchyma and capsule accommodates to sustained expansion of tubular volume.⁸

The early reductions in RBF and GFR of rats infused with furosemide are specific for the kidney, unchanged by volume repletion, renal denervation or blockade of AT1 (angiotensin type 1), vasopressin V1A receptors, or TGF and therefore are likely due predominantly to passive compression of renal vessels.^{8,58–60} The delayed reductions in RBF and GFR persist largely unchanged by volume replacement but are attenuated by blockade of ATIRs, alpha adrenoreceptors, or cyclooxygenase, or by renal nerve deafferentation and therefore entail renal neurohumoral activation.^{4,8,10}

The GFR falls below baseline by 90 minutes after oral administration of loop diuretics to normal human subjects, corresponding to its maximal diuresis and effect on the renal tubules. However, it remains below baseline for at least 24 hours, well after the diuresis has dissipated.²⁶ The mechanism for the delayed reduction in GFR in humans has not been investigated but suggests either a response to neurohumoral activation or to reactive oxygen species induced by ischemia/reperfusion.⁴⁶ Reactive oxygen species could also derive from compressed renal tubules that can be damaged by increases in intrarenal pressure >10 mmHg for 10 minutes.^{61,62} Studies are required to evaluate the mechanism of the delayed and prolonged fall in GFR with rapidly acting loop diuretics. The kidney accommodates over time to sustained increases in intrarenal volume or pressure with diminishing renal vasoconstriction.⁶² Rapidly acting loop diuretics that cause abrupt tubular distension may overwhelm this protective response, whereas diuretics with gradual onset might allow the kidney to accommodate the rise in pressure. This could explain why extended-release torsemide given to normal subjects preserves GFR, while immediate-release formulations reduce it, although this hypothesis requires direct testing.²⁶

CONCLUSIONS

Loop diuretics generally reduce the GFR by >20%, primarily by reducing glomerular plasma flow and LpA despite their engagement of 3 vasodilatory mechanisms:

blockade of TGF and NKCC1 and reduced myogenic tone. These vasodilatory effects are overwhelmed by powerful vasoconstrictor mechanisms that include both passive obstruction (compression of intrarenal capillaries and veins by elevated renal tissue pressure) and active vasoconstriction (driven by intrarenal Ang II, excitatory intrarenal reno-renal reflexes, and release of TxA_2 and vasoconstrictor prostaglandins). The fall in GFR persists well after the tubular effects of the diuretic have dissipated, possibly reflecting effects of increased reactive oxygen species triggered by tubular ischemia and injury from compression.

Crucially, the intrarenal mechanisms of vasoconstriction, like those of postdiuretic salt retention, operate independently of systemic volume depletion.⁶³ Systemic neurohormonal activation likely adds to the delayed intrarenal mechanisms, especially in patients who remain volume overloaded. By reducing RBF, these mechanisms limit diuretic delivery to the tubules, and by reducing GFR, limit natriuresis, thereby contributing to diuretic resistance and explaining why some patients remain poorly responsive to a loop diuretic despite aggressive therapy.



Clinical Perspective

This review illustrates the need for alternative strategies for managing hypervolemia in patients who develop WRF during loop diuretic therapy. The dose of loop diuretics might be reduced by concurrent administration of a sodium-glucose cotransporter type-2 inhibitor, because these drugs synergize with loop diuretics to increase natriuresis.^{64,65} However, sodium-glucose cotransporter type-2 inhibitors also inhibit tubular fluid reabsorption, which may add to the increased intrarenal pressure with loop diuretics.⁶⁶ A mineralocorticoid receptor antagonist may also synergize with loop diuretics in HF. Although it lowers GFR initially, unlike loop diuretics, this fall entails enhanced TGF.^{67,68} The aquaretic tolvaptan has been recommended when GFR is reduced with diuretics.⁶⁹ An extended-release formulation of torsemide has a more gradual but prolonged diuresis and, unlike immediate-release torsemide, does not reduce the GFR.²⁶ Reducing intrarenal pressure directly by 30 mmHg with an experimental JuxtaFlow device (that can apply suction and negative pressure) inserted into the renal pelvis of a pig model of cardiac tamponade improves renal hemodynamics and reverses diuretic resistance rapidly, but this is experimental.⁷⁰ Direct sodium removal in HF using peritoneal dialysis with a sodium-free solution restores diuretic sensitivity and responsiveness that persisted for months after completing treatment, but effects on GFR require study.⁷¹ Ultrafiltration has met with uncertain results.⁷² These available and developing strategies require careful evaluation to establish their relative merits and long-term effects on kidney function.

ARTICLE INFORMATION

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